

Optimal Number of Stages

The following analyses are computed under the validity of Tsiolkovsky's equation and assuming a constant thrust profile.

Step 1: optimal number of stages with k_s assigned globally

The aim of this step is to compute and show the maximum total payload ratio λ_{tot} of a multistage rocket with identical stages ($c_i = c$ and $k_{s_i} = k_s$) as a function of the number of stages N . In this case the structural coefficient is fixed, so this first step can be referred to the **static structural coefficient analysis**. The Tsiolkovsky's equation for the i -th subrocket is:

$$\Delta V_i = -c_i \ln MR_i \quad \text{for } i=1, \dots, N \quad \text{where} \quad c_i = g_0 I_{sp_i} = c \quad \text{and} \quad MR_i = \frac{m_{f_i}}{m_{0_i}}$$

For the identical stages hypothesis it is valid the equal partitioning condition which leads to:

$$\lambda_i = \lambda_{tot}^{\frac{1}{N}} = \lambda \quad \Delta V_i = \frac{\Delta V_{tot}}{N} \quad k_{s_i} = k_s$$

so the final expression for the maximum overall payload ratio is obtained from the solution of a constrained optimization problem by applying the method of Lagrange multipliers:

$$\lambda_{tot_{max}} = \left[\frac{\exp(-\frac{\Delta V_{tot}}{c N}) - k_s}{1 - k_s} \right]^N \quad (1)$$

The plots for different reduced velocity as a function of the assigned fixed structural coefficients and number of stages are shown in figure 1.

In this case, where static structural coefficient has been considered, when there are simple mission requirements characterized by a reduced velocity of 0.5 and 1.0, showed in fig.1(a) and fig. 1(b), the single stage launcher is always able to guarantee a positive and good payload ratio which increases as the structural coefficient decreases. Moreover, the improvement provided by increasing the number of stages is almost negligible because the curves are quite flat.

From fig.1(c) to the end, the trend is different, mostly for the lower number of stages. In fact, in the case of $\Delta V/c = 2.0$, the single stage launcher can reach the required performance only with the best structural coefficient $k_s = 0.10$. This is due to the higher reduced velocity which can be read as an harder requirement for the orbit. The trend of the curves as a function of the number of stages is characterized by a high slope at the beginning, for lower number of stages, and then it decreases as the number of stages increases.

At the end, the last two figures, fig.1(d) and fig. 1(e), are characterized by a reduced velocity of 3.0 and 4.0 which are realistic respectively for a LEO and a lunar transfer application. In this case the advantage derived from increasing the number of stages is evident and it is more and more

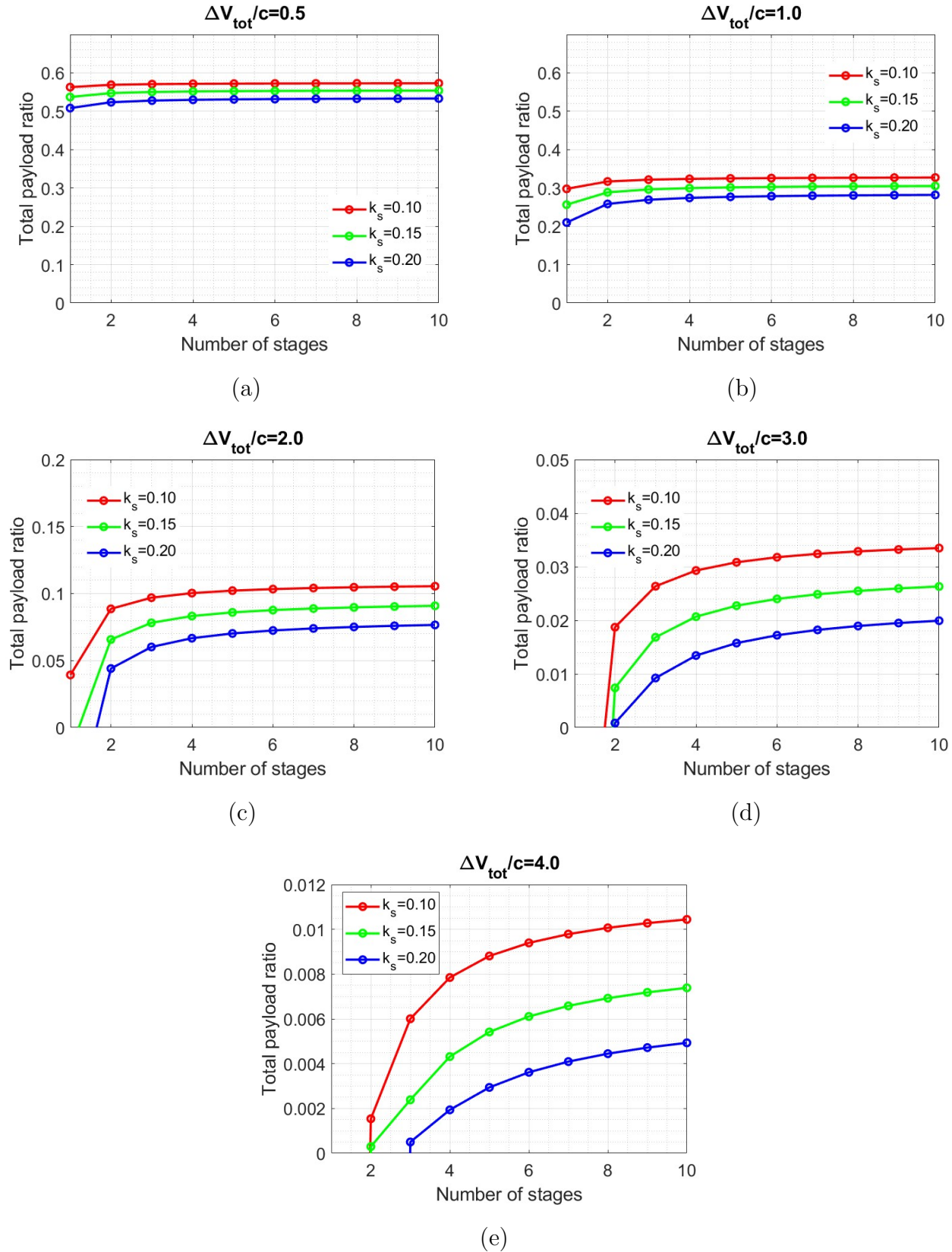


Figure 1: Optimal total payload ratios, $\lambda_{tot_{max}}$, as a function of the number of stages, N , for different combinations of dimensionless ideal velocity and structural coefficient.

important as N increases because of the hardest requirement among all the cases.

In general one can learn that a single stage launcher is not allowed to reach the orbit because the considered values of structural coefficient, which cover a realistic range, is too high to have a positive payload ratio, which is the starting goal of each launcher. Moreover the trend as a function of the structural coefficient is always the same, as explained in the first case, but one can notice that by decreasing k_s the curves become steeper: the improvement of the total payload ratio as a function of the number of stages is higher for better structural distribution (lower structural coefficient).

At the end, in case of static structural coefficient, the total payload ratio always increases as the number of stages increases going towards an asymptotic value which can be evaluated from eq.1 for $N \rightarrow \infty$.

Step 2: optimal number of stages with k_s assigned from individual structural masses

Up to this time it has been considered a fixed k_s , i.e. each stage with the same structural coefficient and it corresponds to an increase of the maximum payload ratio as a function of the number of stages.

In a real case, when one increases the number of stages, the structural coefficient of each stage decreases. This is because in the structural mass of each stage they are hidden two contributes from the tank mass and engine mass. Considering always the condition of identical stages (i.e. each parameter indicated with "i", meaning the validity for the "i-th" stage, is replaced with the parameter without the subscript), the structural mass is now defined as:

$$m_s = m_t + m_e = k_t m_p + k_e m_0$$

where k_t is the mass tank proportionality factor, which indicates the relation between tank mass and propellant mass, and k_e is the engine proportionality factor, which comes from the relation between engine mass and thrust:

$$m_e = \frac{F}{\left(\frac{F}{W}\right)_e g_0} = \frac{m_0 a_0}{g_0 \left(\frac{F}{W}\right)_e} = k_e g_0 \quad \text{where} \quad k_e = \frac{\beta_0}{\left(\frac{F}{W}\right)_e}$$

Moreover it has been introduced: the thrust over weight ratio of the engine $\frac{F}{W}$; the ratio between initial acceleration and gravitational acceleration β_0 .

The new dynamic structural coefficient is given by:

$$k_{s_i} = \frac{m_{s_i}}{m_{p_i} + m_{s_i}} = \frac{k_{t_i}(1 - MR_i) + k_{e_i}}{(1 + k_{t_i})(1 - MR_i) + k_{e_i}} \quad (2)$$

where, for this particular case, the only quantity that varies with the number of stages is the mass ratio MR , which is also the responsible for the variation of the structural coefficient as a function of N .

The expression for the total payload ratio is the same as eq.1 but with the structural coefficient computed in the eq.2.

The parametric analysis is performed for different assigned values of reduced velocity $\frac{\Delta V}{c}$, mass tank proportionality factor k_t , thrust to weight ratio $\frac{F}{W}$ and β_0 . The results are shown in figure 2.

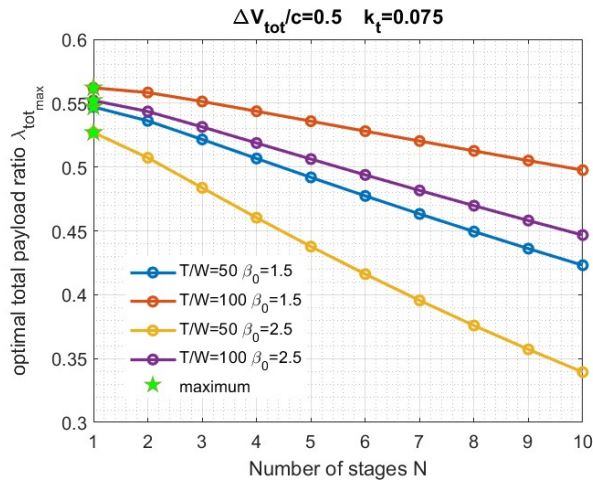
In this case of **dynamic structural coefficient analysis** the structural efficiencies will increase with increasing N and so the optimal number of stages will be finite. This is due to the fact that now one has considered also the contribution of the structural mass of the engines, which increases as the number of the stages increases, and the resulting maximum value for the payload ratio corresponds to a number of stages which is a compromise between the improvement given by increasing the number of stages and the worsening of k_{s_i} for higher N. Another general result is that if $N \rightarrow \infty$ the mass ratio $MR_i \rightarrow 1$ and the value of the total payload ratio tends to zero.

For the first two cases, fig.2(a) and fig.2(b), which correspond to the simplest mission requirements, the maxima correspond to a single-stage solution because increasing N also the mass of the engines increases while the tank mass and propellant mass are low because of the contained reduced velocity.

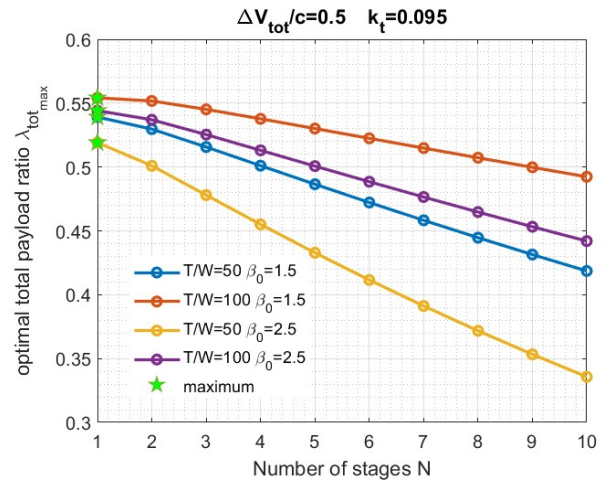
For harder requirements, showed from fig.2(c) to fig.2(d) the behaviour shifts towards an optimal payload ratio for higher number of stages. The reason is that the growth of k_s with N is less rapid with respect to the case of lower mission requirements because an higher reduced velocity implies a larger presence of propellant mass and a consequently lower importance of the engine mass contribution.

The principal results in terms of the given quantities that can be explained are:

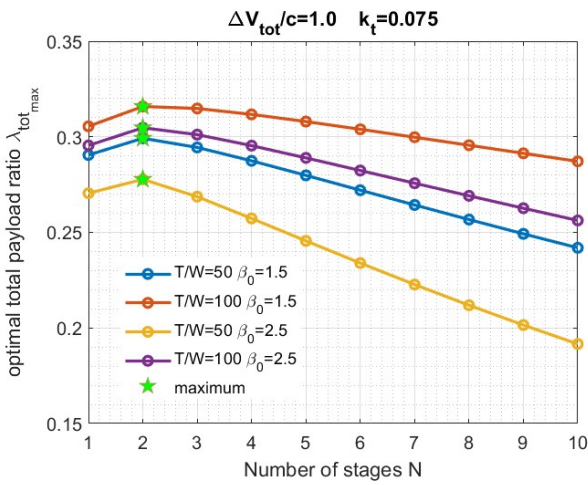
- increasing k_t means an increase of k_s and an obvious decrease of the performances which reflects on a lower $\lambda_{tot_{max}}$;
- an increase of the thrust to weight ratio leads to a decrease of k_e and an opposite behaviour with respect to the previous case, with an increased total payload ratio;
- increasing β_0 means an increase of k_e and a worse performance;
- a more expensive mission in terms of ΔV_{tot} or a less efficient system from the propulsive or structural point of view (lower c or higher k_s) requires a higher number of stages.



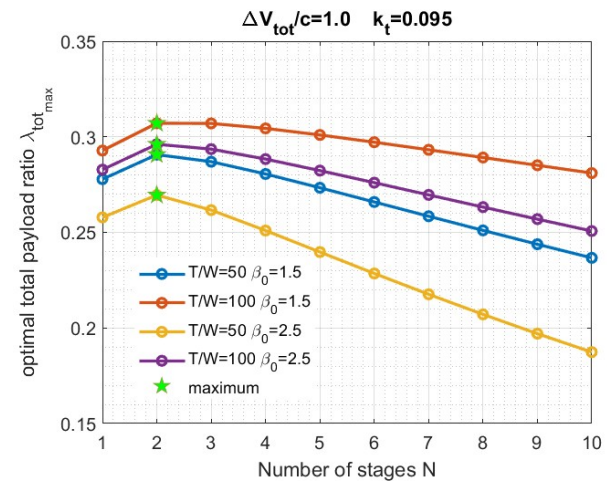
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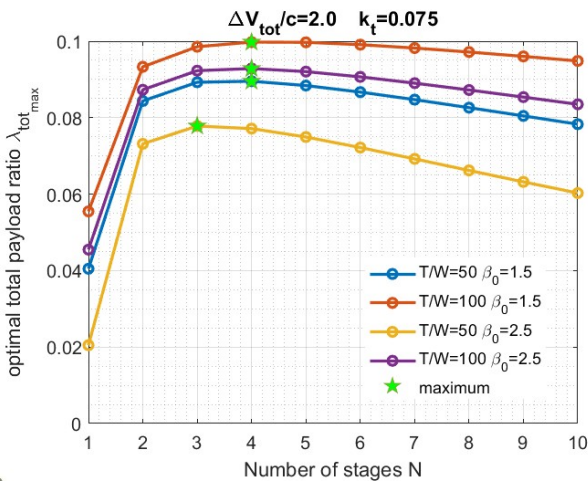
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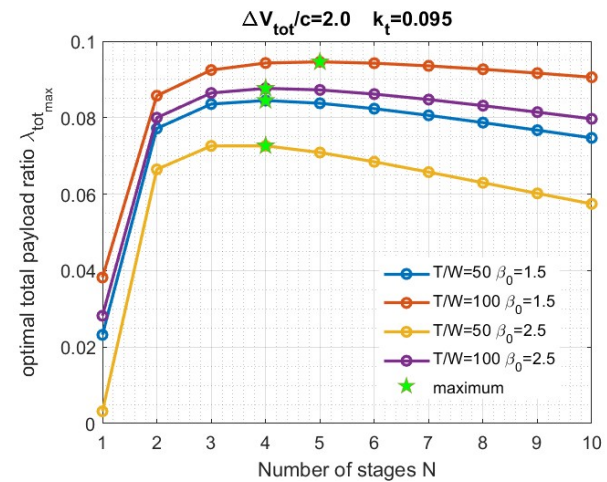
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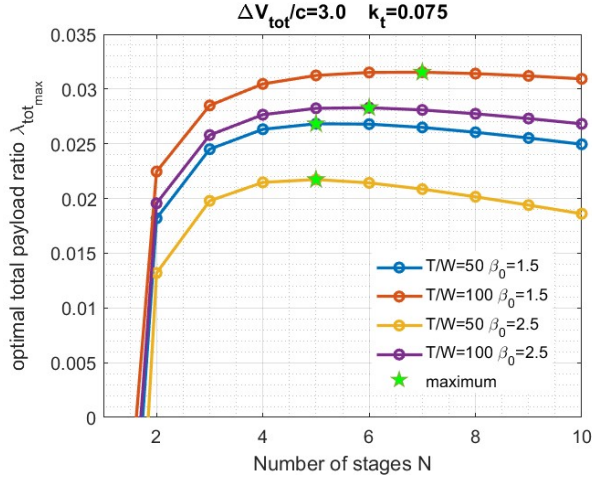
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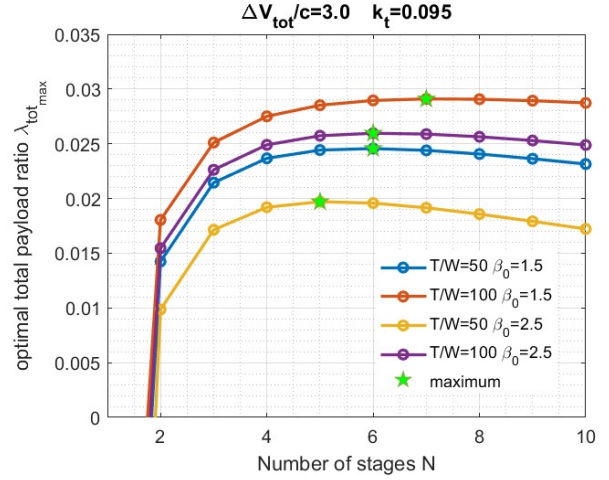
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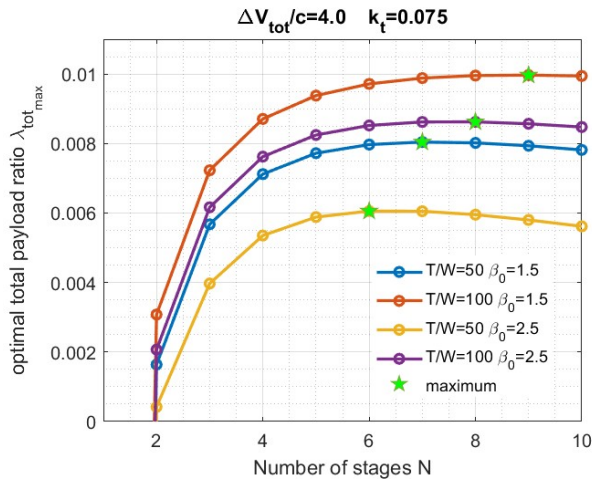
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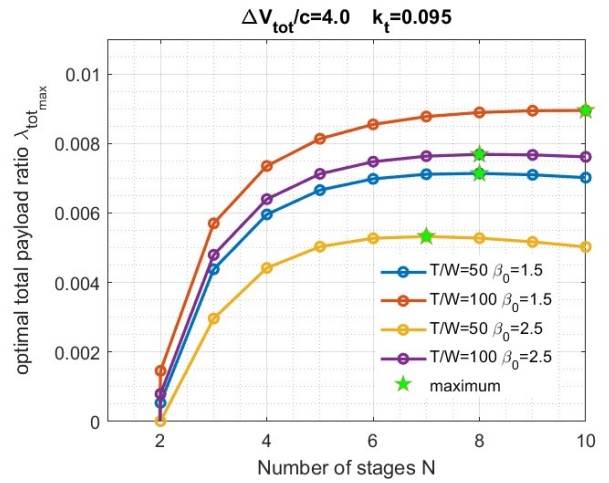
(g)



(h)



(i)



(j)

Figure 2: Optimal total payload ratios, λ_{tot_max} , as a function of the number of stages, N , for different combinations of dimensionless ideal velocity, tank mass factor, thrust to weight ratio and β_0 .

Step 3: optimal number of stages including the effect of gravitational losses

From the previous analysis it is clear that a larger initial acceleration β_0 corresponds to a worse payload ratio, regardless of the number of stages. In reality this is different because one must take into account all type of losses in the velocity increment that are going to decrease the total ΔV . Since the biggest loss is the gravitational one during the vertical path of a launcher, it is necessary to modify the Tsiolkovsky's equation, based on the ideal assumption of zero losses, in order to include also the gravity force and not only thrust.

Starting from the modified equation of motion:

$$m \frac{dV}{dt} = F - mg_0$$

with a series of mathematical steps one finally obtain the velocity increment of each sub-rocket:

$$\Delta V_i = c_i \ln MR_i + \frac{g_0}{F_i} c_i (m_{f_i} - m_{0_i}) = c_i \left[\ln MR_i + \frac{1}{\beta_0} (1 - MR_i) \right] \quad (3)$$

where effective exhaust velocity and the thrust are considered as constant during the flight.

The **gravitational losses model analysis** is done considering the same assumption of the previous step i.e. same values for the constant quantities and the assumption of identical stages, which leads to the new expression for the reduced velocity of each stage:

$$\frac{\Delta V_{tot}}{cN} = -\ln MR - \frac{1}{\beta_0} (1 - MR)$$

and this is solved as a non linear equation in order to evaluate the mass ratio and substitute this in the eq.1 for the maximum total payload ratio. All the results are shown in fig.3.

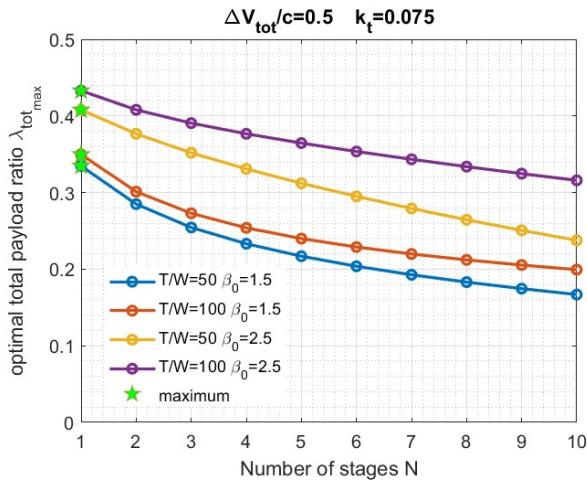
The trend of the curves are the same of the previous step, especially for the considerations about the best number of stages. The role of gravitational losses is that of shifts all curves to a lower optimal total payload ratio.

Moreover, from eq.3, it is possible to see that ah higher specific impulse c results on bigger losses because it means same thrust applied for a long time and so bigger gravitational losses.

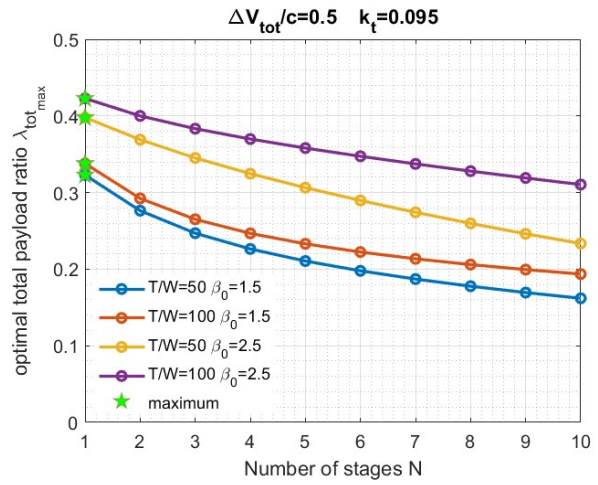
It is also instructive to analyze the ratio between velocity increment derived by gravitational losses and the one derived from the ideal case. From eq.3 the term relative to losses is:

$$\Delta V_{loss} = \frac{1 - MR}{\beta_0} c$$

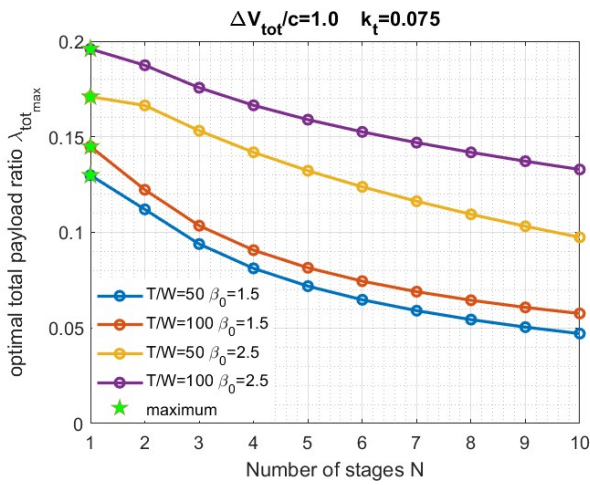
so the importance of losses with respect to the ideal ΔV can be evaluated as:



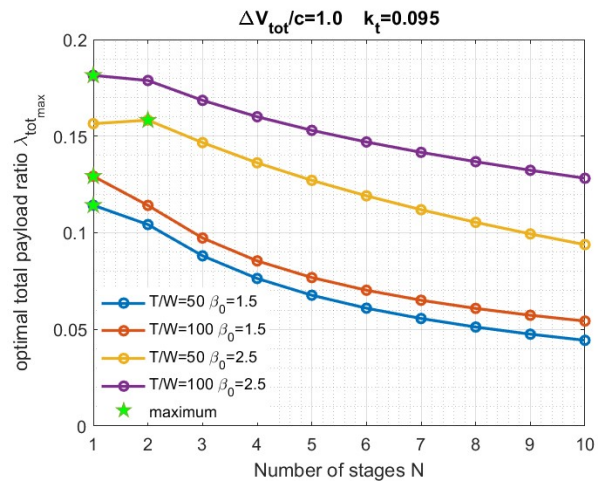
(a)



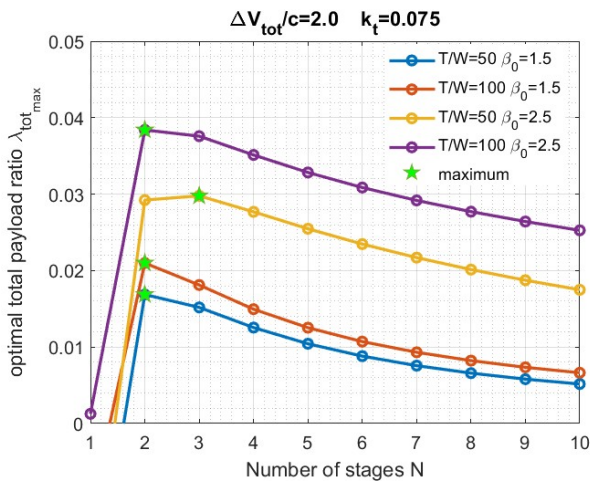
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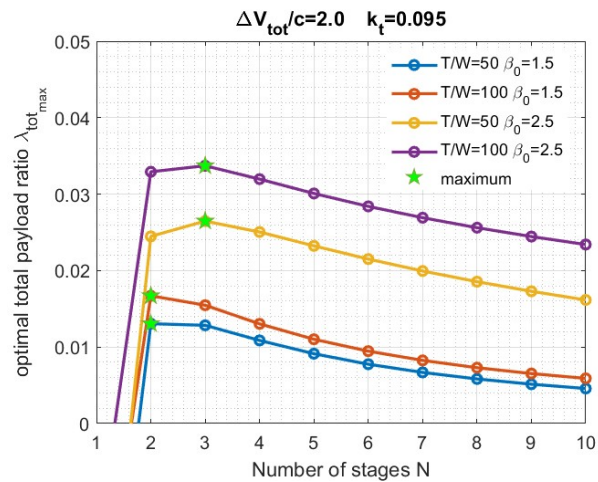
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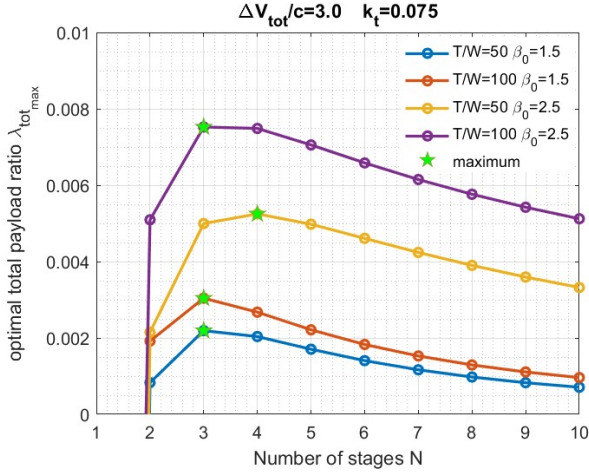
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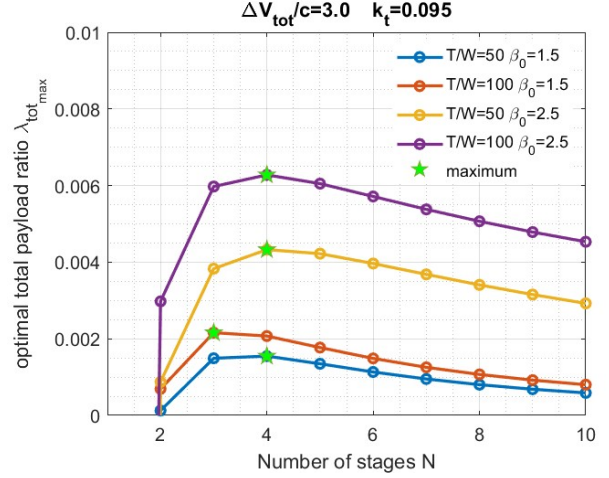
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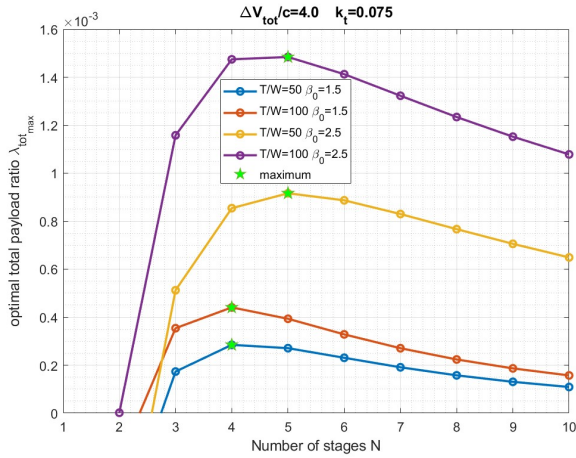
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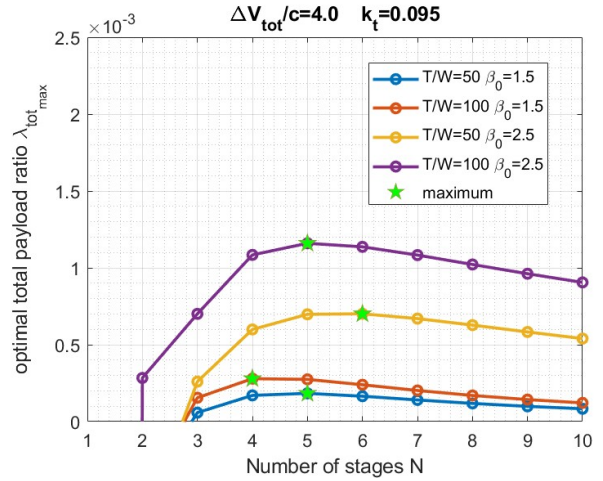
(g)



(h)



(i)



(j)

Figure 3: Optimal total payload ratios, $\lambda_{tot\max}$, as a function of the number of stages, N , for different combinations of dimensionless ideal velocity, tank mass factor, thrust to weight ratio and β_0 , taking into account gravitational losses.

$$\frac{\Delta V_{loss}}{\Delta V_{ideal}} = \frac{1}{-\ln MR} \frac{1 - MR}{\beta_0}$$

From this expression it is clear that losses can be reduced by reducing the mass ratio and it is demonstrated in fig.4. Another parameter that could be varied is β_0 , in fact higher values corresponds to minor losses, as it can be shown in fig.5, where it is clear also the behaviour in terms of different number of stages: larger N implies a larger $\Delta V_{loss}/c$.

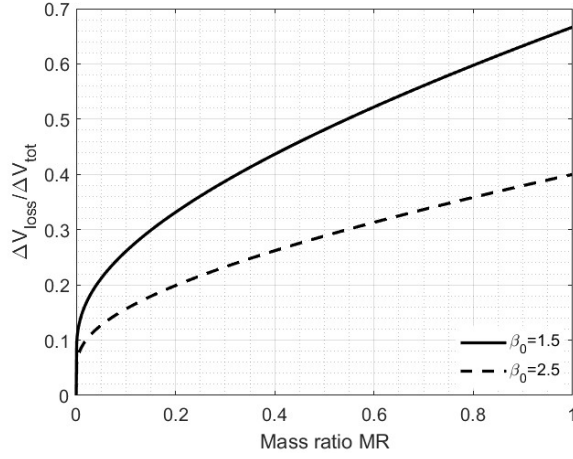


Figure 4: Losses relative importance with respect to the ideal ΔV , as a function of MR, for different values of β_0 .

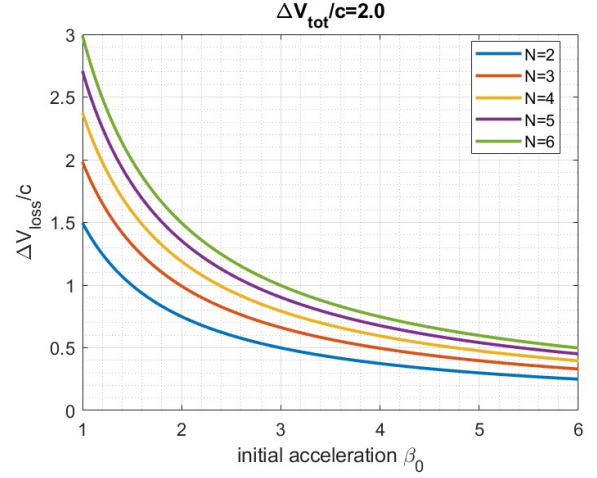


Figure 5: Dimensionless gravity losses as a function of β_0 and number of stages.

The overall payload ratio of a launcher as a function of the initial acceleration and number of stages is shown in fig.6. One can notice that there is an optimum β_0 that maximizes $\lambda_{tot_{max}}$ for each N and the plot depends on the specific mission requirements. In fact, increasing β_0 the losses are reduced but the k_e coefficient increases and it results on a worse structural coefficient.

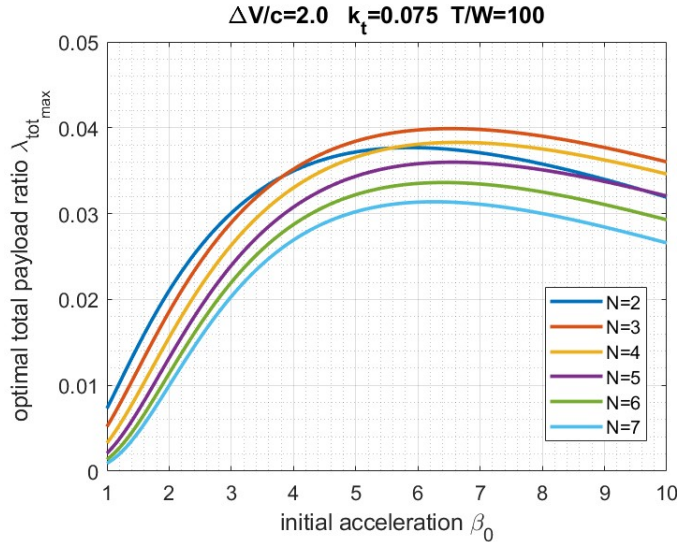


Figure 6: Total payload ratio as a function of the initial acceleration factor and number of stages for a rocket with identical stages.

A final overview of the present study can be done in order to compare the three cases:

- The analysis based on the *fixed structural coefficient* shows an asymptotic increasing behaviour of the maximum total payload ratio as a function of N , which corresponds to an infinite number of stages as the best solution.
- The analysis based on the *dynamic structural coefficient* shows a maximum payload ratio for a finite number of stages, which is of course a direct consequence of a more realistic approach with respect to the previous one.
- The analysis based on the gravitational model is the one for which the $\lambda_{tot_{max}}$ is reached for the lower number of stages because of the role played by the gravitational losses on its evaluation.

The real case is a compromise between dynamic structural coefficient and the analysis that takes into account gravitational losses.